

GANESH GANTI

EMBEDDED SYSTEMS & SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Embedded systems engineer with 3+ years of professional experience spanning high-precision semiconductor systems, embedded Linux, RF-connected devices, production test automation, and secure communications. Currently working through ALTEN Nederland at ASML in State Positioning & Measurement, with responsibilities evolving from software development toward system integration. MSc Embedded Systems from TU Eindhoven with thesis grade 9.0/10 and research in privacy-preserving drone Remote-ID. Strong foundation in C/C++, Python, Linux/RTOS, hardware-software integration, debugging, and communication protocols.

EXPERIENCE

Consultant Technology - Software Development /

2025 - Present

Integration | ALTEN Nederland (assignment at ASML - State Positioning & Measurement)

Eindhoven / Veldhoven, Netherlands

- Contribute to software development and integration within ASML's State Positioning & Measurement domain, part of a high-precision metrology environment.
- Work across software and subsystem boundaries to investigate integration behavior, resolve technical issues, and support verification of system-level functionality.
- Progressed from a development-focused role toward integration, increasing emphasis on interfaces, end-to-end behavior, robustness, and multidisciplinary problem solving.

Embedded Systems Engineer | Rocateq International B.V.

2023 - 2024

Rotterdam, Netherlands

- Developed an IoT application on single-board embedded Linux systems using Python and MQTT for real-time data processing and device communication.
- Designed and implemented a custom security protocol in C++ on Zephyr RTOS for RF broadcasts, protecting system transceivers against replay attacks.
- Led development of BLE over-the-air firmware update functionality for transceivers and introduced power-management improvements to reduce energy use.

Embedded Test Engineer | Prodrive Technologies B.V.

2022 - 2023

Eindhoven, Netherlands

- Designed and implemented C++-based embedded-system tests for in-house product manufacturing and test equipment.
- Supported reliable automated testing across multiple manufactured products and studied the use of IoT within product manufacturing test development.

Technical Documentation Engineer (Part-time) |

2019 - 2020

M5Stack

Remote

- Worked with the engineering team to review and improve product requirements, technical specifications, user manuals, and product documentation.

IoT Intern | Fluentgrid Pvt. Ltd.

2019

Visakhapatnam, India

- Developed an energy-efficient smart-streetlight prototype using proximity sensing and cameras for presence-aware control; supported testing and documentation.

EDUCATION

MSc Embedded Systems | Eindhoven University of Technology (TU/e)
Eindhoven, Netherlands

2021 - 2023

- Master's thesis: Confidential Pilot Location Disclosure in Drone Remote-ID - thesis grade 9.0/10.
- Thesis work published in PoPETs and presented at the 2024 PETS Symposium.

BTech Electronics and Communication Engineering |

Vellore Institute of Technology
Vellore, India

2016 - 2020

- CGPA: 9.03/10.

RESEARCH & SELECTED TECHNICAL WORK

Confidential Pilot Location Disclosure in Drone Remote-ID (MSc Thesis)

- Developed a privacy-preserving Remote-ID approach for controlled disclosure of pilot location while maintaining authenticity and regulatory compatibility.
- Implemented applied cryptographic mechanisms in C++ on ESP32-class hardware and worked at the 802.11 MAC / wireless-protocol boundary.
- Combined embedded security, cryptography, wireless communications, and constrained-system implementation.

Recent embedded / IoT projects

- ESP32 touch Pomodoro / clock device - 240x240 touch UI, NTP time, Wi-Fi configuration, inactivity handling, and persistent session state.
- Indoor air-quality monitor - ENS160 + AHT21 sensing integrated with MQTT and Home Assistant.
- ESP32 BLE input device - Bluetooth LE mouse-style control for iPhone.

Earlier selected projects

- Digitalizing Pen - MEMS-based smart pen with LSTM-based character recognition for digitizing handwritten content.
- Frame-IoT & ZeetOS - IoT framework and lightweight real-time operating-system work for protocol integration and embedded deployment.

TECHNICAL SKILLS

Programming: C, C++, Python, Rust; exposure to VHDL, JavaScript and Assembly

Embedded / OS: Embedded Linux, Yocto Project, FreeRTOS, Zephyr RTOS; ARM/ST, Espressif, Nordic Semiconductor, Raspberry Pi, BeagleBoard

Interfaces & Connectivity: UART, SPI, I2C, USB, MQTT, CoAP, Wi-Fi, Bluetooth LE, LoRaWAN, IEEE 802.15.4, Thread

Debugging / Development: JTAG, Git, Linux, Agile development; hardware-software integration and embedded-system test

Security: Secure coding, cryptographic algorithms, embedded communication security, TLS/SSH

Engineering Tools: MATLAB, LTspice, GNU Radio, Keil, Cadence, Proteus, Node-RED

LEADERSHIP & ADDITIONAL

Academic / leadership: Student member, TU/e Embedded Systems Curriculum Committee (2023); teaching assistant experience at TU/e and VIT; former IEEE Robotics & Automation Society student-chapter manager/project head.

Competitions: 1st place, IoT Hackathon (2019); winner, IEEE Makeathon (2018).

Languages: Telugu (native), English (fluent), Hindi (fluent), Dutch (A2).

Amateur Radio: HAREC-qualified licensed radio amateur; Dutch callsign PG4NTI and UK callsign M1IGT.